Final Exam

Graded

Student

Christian Patrick Byrne

Total Points

96 / 100 pts

Question 1

1 **32** / 32 pts

✔ **– 0 pts** Correct

– 2 pts a) wrong conclusion

– 1 pt a) in general relevant to the topics.

– 2 pts a) wrong or missing explanation

– 2 pts b) wrong conclusion

– 1 pt b) in general relevant to the topics.

– 2 pts b) wrong or missing explanation

– 2 pts c) wrong conclusion

– 1 pt c) in general relevant to the topics.

– 2 pts c) wrong or missing explanation

– 2 pts d) wrong conclusion

– 1 pt d) in general relevant to the topics.

– 2 pts d) wrong or missing explanation

– 2 pts e) wrong conclusion

– 1 pt e) in general relevant to the topics.

– 2 pts e) wrong or missing explanation

– 2 pts f) wrong conclusion

– 1 pt f) in general relevant to the topics.

– 2 pts f) wrong or missing explanation

– 2 pts g) wrong conclusion

– 1 pt g) in general relevant to the topics.

– 2 pts g) wrong or missing explanation

– 2 pts h) wrong conclusion

– 1 pt h) in general relevant to the topics.

– 2 pts h) wrong or missing explanation

Question 2

2 **8** / 8 pts

- ✔ **− 0 pts** Correct
- **− 2 pts** (a). incorrect prediction for sales
- **− 2 pts** (b). incorrect calculation for the confidence interval
- **− 4 pts** (b). solution or details largely missing
- **− 4 pts** (a). solution or details largely missing

Question 3

3 **5** / 5 pts

- ✔ **− 0 pts** Correct
- **− 3 pts** incorrect definition for logistic model
- **− 4 pts** solution largely missing
- **− 2 pts** mirror mistake
- **− 5 pts** solution missing

Question 4

4 **14** / 16 pts

- **− 0 pts** Correct
- ✔ **− 1 pt** (a) mostly correct method, wrong final result
- **− 2 pts** (a)(mostly) wrong method but make effort
- **− 4 pts** (a)not submitted/no effort
- **− 1 pt** (b) mostly correct method, wrong final result
- **− 2 pts** (b)(mostly) wrong method but make effort
- **− 4 pts** (b)not submitted/no effort
- **− 1 pt** (c) mostly correct method, wrong final result , not report both p and the expected evaluation score
- **− 2 pts** (c)(mostly) wrong method but make effort
- **− 4 pts** (c)not submitted/no effort
- ✔ **− 1 pt** (d) mostly correct method, wrong final result
- **− 2 pts** (d)(mostly) wrong method but make effort
- **− 4 pts** (d)not submitted/no effort

Question 5

5 — **8** / 8 pts

✔ **– 0 pts** Correct

– 2 pts (b). the new classifier could change depending on the new training set

– 2 pts (a). it is not always possible

– 4 pts (a). missing detailed discussion

– 4 pts (b). missing detailed discussion

Question 6

6 — **5** / 5 pts

✔ **– 0 pts** Correct

– 3 pts It is possible

– 2 pts fail to provide such an instance

– 5 pts solution missing

Question 7

7 — **8** / 8 pts

✔ **– 0 pts** Correct

– 2 pts (a). the problem statement requires to show the output clustering tree. No tree is given here.

– 2 pts (b). the problem statement requires to show the output clustering tree. No tree is given here.

– 4 pts (a). solution / details largely missing

– 4 pts (b). solution / details largely missing

– 2 pts (a). incorrect instance given

– 2 pts (b). incorrect instance given

– 2 pts (a) incomplete tree

– 2 pts (b) incomplete tree

Question 8

8 — **3** / 5 pts

– 0 pts Correct

– 3 pts incorrect calculation, it is a beach day

✔ **– 2 pts** need to calculate for both beach and non beach day to compare

– 5 pts detailed solution / calculation missing

– 4 pts detailed calculation missing

Question 9

9 **5** / 5 pts

✔ **− 0 pts** Correct

− 3 pts insufficient explanation and intuition

− 5 pts solution missing

Question 10

10 **8** / 8 pts

✔ **− 0 pts** Correct

− 2 pts (a). incorrect / incomplete calculation for Gini index

− 2 pts (b). the decision tree will do the second split.

− 2 pts (b). incorrect / incomplete calculation for Gini index

− 4 pts (b). solution / calculation details largely missing

− 4 pts (a). solution / calculation details largely missing

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CSC 380 Final Exam

- *Exam Duration: 120 minutes.*
- *There are 10 questions.*
- *You can make use of a hand-written cheat sheet (double-sided letter size).*
- *Provide your answers* **only inside the boxes** *underneath the questions.*
- *Provide neat and concise solutions.*
- *Good luck!*

Question 1 [32 pts]: Mark **True** or **False**, and explain briefly with one or two sentences. Note that you will only receive partial credit if your answer is correct, but your explanation is not or if your explanation is missing.

(a) Given a data matrix A where the rows correspond to the items and the columns correspond to the features, $A^T A$ measures the *in-syncness* of the features.

TRUE **FALSE**

A^T transposes the the matrix such that columns (features) become rows — thus, A^TA is finding dot products of rows (which are features) of A^T with columns of A (which are also features). Since dot product is a measure of in-syncness, A^TA thus measures in-syncness of features.

(b) The *precision* of a binary classifier is the ratio of True Positives to the total number of Positives in the data set.

TRUE **FALSE**

The metric which measures the ratio of true positives to all positives in the data set is recall, not precision. Precision is ratio of correct positive predictions to all positive predictions — not all positive instances in the data set.

(c) The value of R^2 is always nonnegative.

TRUE **FALSE**

R^2 measures variability of y explained using x: $R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (y_i - \bar{y})^2}$. Consider the example wherein there is a larger difference between (predicted values and actual values) than there is between (actual values and the mean $\bar{y}$). Then, the fraction is >1 and we have $R^2 = 1 -$ val greater than 1.

(d) In stochastic gradient descent we see each data point exactly once in one training epoch.

TRUE **FALSE**

In one epoch of stochastic gradient descent, we iteratively update weights after computing slope of RSS curve and we do it for each data point one time until the epoch is concluded.

(**Continue Question 1**)

(e) If we are doing linear regression we should use *ridge*, whereas if we are doing logistic regression we should use *LASSO* as our regularization technique.

TRUE **FALSE**

Although you may make arguments that parts of that statement are true or at least true in certain scenarios, the idea that we should always use ridge when doing linear is enough to make entire statement false. There are certainly scenarios where ridge yields better linear regression results — e.g., if some weights always need to be ≥ 0.

(f) The top and the bottom whiskers indicate respectively the third and the first quartile in a box plot.

TRUE **FALSE**

The box's edges represent the 1st and 3rd quartile — not the whiskers. The whiskers represent the outlier bounds — that is, Q1 − 1.5IQR and Q3 + 1.5IQR.

(g) If we use soft margins in a SVM, there will be no training observations that are classified incorrectly.

TRUE **FALSE**

In fact, using soft margins will allow some observations to be classified incorrectly — as it affords our hyperplane some slack such that some relatively problematic or hard-to-classify training data points can be on the wrong side of the hyperplane.

(h) R^2 is the same as the square of correlation coefficient in simple linear regression.

TRUE **FALSE**

R^2 is a generalization of the same concept represented by the square of the correlation coefficient in SLR — they measure the fraction of variability of y explained by x (i.e, proportion of variance in in dependent variable predictable from the independent variable).

Question 2 [8 pts]: Suppose we have a data set with three predictors, $X_1 = TV, X_2 = radio$, and $X_3 =$ Interaction between *TV* and *radio*. Here *TV* and *radio*, indicate respectively the budget spent on TV and radio advertising. The response is *sales*, the number of units sold. Suppose we use least squares to fit a linear regression model, and get $\hat{w}_0 = 6.75, \hat{w}_1 = 0.02, \hat{w}_2 = 0.028, \hat{w}_3 = 0.001$.

(a) Predict the sales of a market spending \$2000 on TV and \$1000 on radio advertisement.

Answer:

$\hat{y} = \hat{\omega}_0 + \hat{\omega}_1 x_1 + \hat{\omega}_2 x_2 + \hat{\omega}_3 x_3 = 6.75 + 0.02(2000) + 0.028(1000) + 0.001(2000+1000)$

assuming X_3 = TV budget + Radio Budget (in dollars)

Alternatively, use function $K(2000, 1000)$ that maps (TV, radio) to expected X_3.

(b) Suppose the standard error associated with w_1 is 0.002. What is the 95% confidence interval for w_1?

Answer:

For 95% confidence interval, use critical value $K = 1.96$, then:

CI is $\omega_1 \pm 1.96 \cdot SE(\omega_1) = \boxed{0.02 \pm 1.96 \cdot 0.002}$

if $K = 2$, then $0.02 \pm 2 \cdot 0.002$

Question 3 [5 pts]: We fit a logistic regression model that classifies the students taking a test with respect to *passing* or *failing* the test on a single feature, the number of hours studied. Let $\hat{P}$ for a data point denote the probability that the data point corresponds to a *pass* as predicted by the classifier. Assume after fitting the model, we got the intercept $w_0 = -0.15$ and the coefficient of the feature number of hours studied $w_1 = 0.575$. What is the probability that the model predicts a new student who has studied for 4 hours passes the test?

Answer:

For logistic regression, $P(y=1|x) = \frac{1}{1+\exp(-(\omega_0+\omega_1 x))} \Rightarrow$

$\hat{P} = P(y=1|4) = \frac{1}{1+e^{(-(-0.15+0.575\cdot 4))}}$

$f \leq \frac{1}{2}$ = pos prop

Question 4 [16 pts]: Suppose $f \leq 1/2$ is the fraction of positive elements in a binary classification data set. For classification we use a random classifier that predicts positive with a certain probability p and predicts negative with probability $1 - p$. What is the probability p that the random classifier should guess positive, as a function of f, in order to maximize each specific evaluation metric below? Report both p and the expected evaluation score the random classifier achieves.

(a) Accuracy.

Answer:

Accuracy $= \frac{TP+TN}{TP+TN+FP+FN}$; correct positive predictions $= f \cdot n \cdot p$, correct negative predictions $= (1-f)(1-p)n$ $\Rightarrow$ accuracy $= \frac{f \cdot n \cdot p + (1-f)(1-p)n}{n} = fp + (1-f)(1-p) = fp - f - p + fp + 1 = 2fp - f - p + 1 \Rightarrow$ we should set $p = f$, yielding score of fp

(b) Precision.

Answer: fn pos, predict $p \rightarrow fnp$ correct, pn all $\rightarrow$ precision $= \frac{fnp}{np} = f$

precision $= \frac{TP}{TP+FP}$ correct positive predictions $= f \cdot n \cdot p$, all positive predictions $= p \cdot n \Rightarrow$ precision $= \frac{f \cdot n \cdot p}{n \cdot p} = f$. Thus, the p-value we choose does not matter — the precision will always be f

(c) Recall.

Answer:

correct positive predictions $= f \cdot n \cdot p$, all actual positive instances $= f \cdot n \Rightarrow$ recall $= \frac{f \cdot n \cdot p}{f \cdot n} = p \Rightarrow$ to maximize recall, we should set p to 1, which should yield a score of 1.

(d) F-score.

Answer:

$f1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = 2 \cdot \frac{f(p)}{f+p}$ (calculated above) $\Rightarrow 2fp = f1 \cdot (f+p) \Rightarrow 2fp = f1 \cdot f + f1 \cdot p$

set $p = 2/(1-f1)$ yielding score of $2\frac{fp}{f+p}$

$2\frac{fp}{f+p} = 1 \Rightarrow$

$\frac{fp}{f+p} = \frac{1}{2} \Rightarrow$

$2f \cdot \frac{1}{2} \Rightarrow 2f = \frac{1}{2}f + 1$ $\frac{1}{3}p$ $\frac{1}{2}$

$f = 2$ $\frac{3}{2}f = 1 \Rightarrow$ $\frac{1}{3} + p$

$f = \frac{2}{3}$

$\frac{f \cdot \text{prec}}{f + \text{prec}} \cdot 2 = f1$

$\frac{f \cdot \text{prec}}{f + \text{prec}} = \frac{f1}{2}$

$f \cdot \text{prec} = \frac{1}{2}f + \frac{1}{2}\text{prec}$

$\frac{1}{2p}$ $\frac{1}{4} + \frac{1}{2p}$

$\frac{1}{3}p = \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2}p$

$\frac{1}{3}p = \frac{1}{6} + \frac{1}{2}p \Rightarrow \frac{1}{6} = -\frac{1}{6}p \Rightarrow -1$

$f1 = 2 \cdot \frac{\text{prec} \cdot \text{recall}}{\text{prec} + \text{recall}}$

$f1 = 2 \cdot \frac{f \cdot \text{recall}}{f + \text{recall}}$

$\frac{f1}{2} = \frac{f \cdot fnp}{fn} \cdot \frac{1}{f+f}$

Question 5 [8 pts]: Suppose we train a maximum margin separator using a training set and obtain a binary classifier C that perfectly separates the two classes. Assume we added one new observation to our training set, retrained a maximum margin separator with the new training set.

(a) Can we always obtain a maximum margin separator that perfectly separates the two classes with the new training set? Discuss why or why not?

Answer:

No - it may be the case that the addition of a single new training data point makes it so the data points are no longer linearly separable.

(b) If it is possible to obtain a maximum margin separator with the new training set, is the new classifier the same as the old one C? Discuss why or why not.

Answer:

NO - the addition of a single new data point can alter the nature of the hyperplane such that the new classifier uses a different equation - with a new hyperplane equation, it may occur that new test data used for inference are classified differently than before.

Question 6 [5 pts]: Suppose a two-class, $k = 3$ nearest-neighbor classifier is trained with at least three positive points and at least three negative points. Could it be possible this classifier labels all new examples as positive? If it is possible provide such a training set, otherwise discuss why this can not be the case.

Answer:

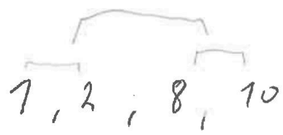
Yes - for instance, imagine that there are many more positive instances than negative such that every cluster is dominated by positive data points - this still satisfies the constraints of the problem (at least three pos, at least three negative points) and will label all new examples as positive.

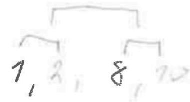
Consider 10 points: that form clusters:

positive: 4, 6, 19, 21, 39, 41
negative: 5, 20, 40

cluster 1: $4_P, 5_N, 6_P$
cluster 2: $19_P, 20_N, 21_P$
cluster 3: $39_P, 40_N, 41_P$

$\Big\}$ all 3 clusters have majority positive data points



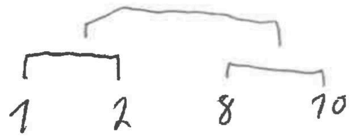


Question 7 [8pts]:

(a) Pick 4 1D points from the set of 10 possible integers from 1 to 10 such that the single-link clustering and the complete-link clustering both give the same hierarchical agglomerative clustering output. Show the output clustering tree.

Answer:

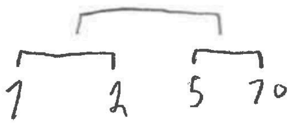
$\{1, 2, 8, 10\}$

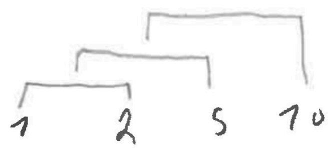


(b) Pick 4 1D points from the set of 10 possible integers from 1 to 10 such that the single-link clustering and the complete-link clustering give different hierarchical agglomerative clustering outputs. Show the two output clustering trees.

Answer:

$\{1, 2, 5, 10\}$





Question 8 [5 pts]: Using the naive Bayes classifier with respect to the following table, decide whether a new day with $Outlook = Cloudy, Temp = High, Humidity = High$ is a beach day.

Day	Outlook	Temp	Humidity	Beach?
1	Sunny	High	High	Yes
2	Sunny	High	Normal	Yes
3	Sunny	Low	Normal	No
4	Sunny	Mild	High	Yes
5	Rain	Mild	Normal	No
6	Rain	High	High	No
7	Rain	Low	Normal	No
8	Cloudy	High	High	No
9	Cloudy	High	Normal	Yes
10	Cloudy	Mild	Normal	No

temp, humid
Cloudy, High, High

$p(b|s,m,h) =$
$p(b)p(s|b)p(m|b)p(h|b)$

Answer:

$p(Beach \mid Sunny, High_{temp}, High_{humid}) = p(Beach) \cdot p(Sunny \mid Beach) \cdot p(High_{temp} \mid Beach) \cdot p(High_{humid} \mid Beach)$

$P(Beach) = \frac{4}{10} = \frac{2}{5}$ $\quad = \frac{2}{5} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{2}$

$P(Sunny \mid Beach) = 3/4$

$P(High_{temp} \mid Beach) = 3/4$ $\quad = \frac{6}{20} \cdot \frac{3}{8} = \frac{18}{160}$

$P(High_{humid} \mid Beach) = 2/4 = 1/2$

Question 9 [5 pts]: How does the random forest technique differ from the bagging technique in the context of decision trees? Discuss the intuition behind this difference.

Answer:

Both random forest and bagging create random samples with replacement of the training data in order to train independent decision trees; however, random forest differs from bagging by also using a random subset of features/inputs for each split. The intuition behind this is that it may be the case that a small subset of feature(s) dominate in terms of predictive power — so each independent tree ultimately classifies all new data points the same way. By using random subsets of features, we potentially allow other features to influence prediction in some trees.

$$15C_1, 15C_2 \xrightarrow{\text{split with } X_1} \begin{array}{l|l} S_t \quad 5C_1, 5C_2 & \text{total } 10 \\ \text{-------- } X_1 & \\ S_b \quad 10C_1, 10C_2 & \text{total } 20 \end{array} \Big\} n=30$$

Question 10 [8 pts]: Suppose we are doing binary classification into two classes C_1, C_2 using the decision tree model. Assume a node u in the tree corresponds to a partition of the data set that has 15 data points of class C_1 and 15 data points of class C_2. Suppose a possible split at u based on a feature X_1 provides the partitions S_t, S_b. The partition S_t has 5 data points of class C_1, 5 data points of class C_2, whereas S_b has 10 data points of class C_1 and 10 data points of class C_2.

(a) Compute the Gini index of this split.

Answer:

$\frac{C_1 \text{ total}}{n} = \frac{15}{30} = \frac{1}{2}$

$\frac{C_2 \text{ total}}{n} = \frac{15}{30} = \frac{1}{2}$

(where Gini index is represented by G)

$$G = \frac{1}{2}\left[1 - \left(\left(\frac{5}{10}\right)^2 + \left(\frac{5}{10}\right)^2\right)\right] + \frac{1}{2}\left[1 - \left(\left(\frac{10}{20}\right)^2 + \left(\frac{10}{20}\right)^2\right)\right]$$
$$= \frac{1}{2}\left[1 - \left(\frac{1}{4} + \frac{1}{4}\right)\right] + \frac{1}{2}\left[1 - \left(\frac{1}{4} + \frac{1}{4}\right)\right]$$
$$= \frac{1}{2}\left[\frac{1}{2}\right] + \frac{1}{2}\left[\frac{1}{2}\right] = \frac{1}{4} + \frac{1}{4} = \boxed{\frac{1}{2}}$$

(b) Assume another split at u based on another feature X_2 provides the partitions S'_t, S'_b such that S'_t has 15 data points of class C_1 and no data points of class C_2, whereas S'_b has 15 data points of class C_2 and no data points of class C_1. Based on the Gini index values of the splits, which split would the decision tree choose to do at u, the one in part (a) above or this one?

Answer:

$$G = \frac{1}{2}\left[1 - \left(\left(\frac{15}{15}\right)^2 + \left(\frac{0}{15}\right)^2\right)\right] + \frac{1}{2}\left[1 - \left(\left(\frac{0}{15}\right)^2 + \left(\frac{15}{15}\right)^2\right)\right]$$
$$= \frac{1}{2}\left[1 - (1)\right] + \frac{1}{2}\left[1 - (1)\right] = \frac{1}{2}[0] + \frac{1}{2}[0] = 0$$

Based on the Gini indices of the splits, the decision tree in part (b) minimizes the Gini index relatively better, so it would be chosen over the other split.

$S'_t \quad 15C_1 \quad 0C_2 \quad$ total 15

$S'_b \quad 15C_2 \quad 0C_2 \quad$ total 15